

CSE420 Assignment 1

Fall 25

Total Marks: 10

1. Convert the following regular expression to a Deterministic Finite Automaton (DFA) using the **direct method**: [8]

$(a|b)^*a(bb^*)a$

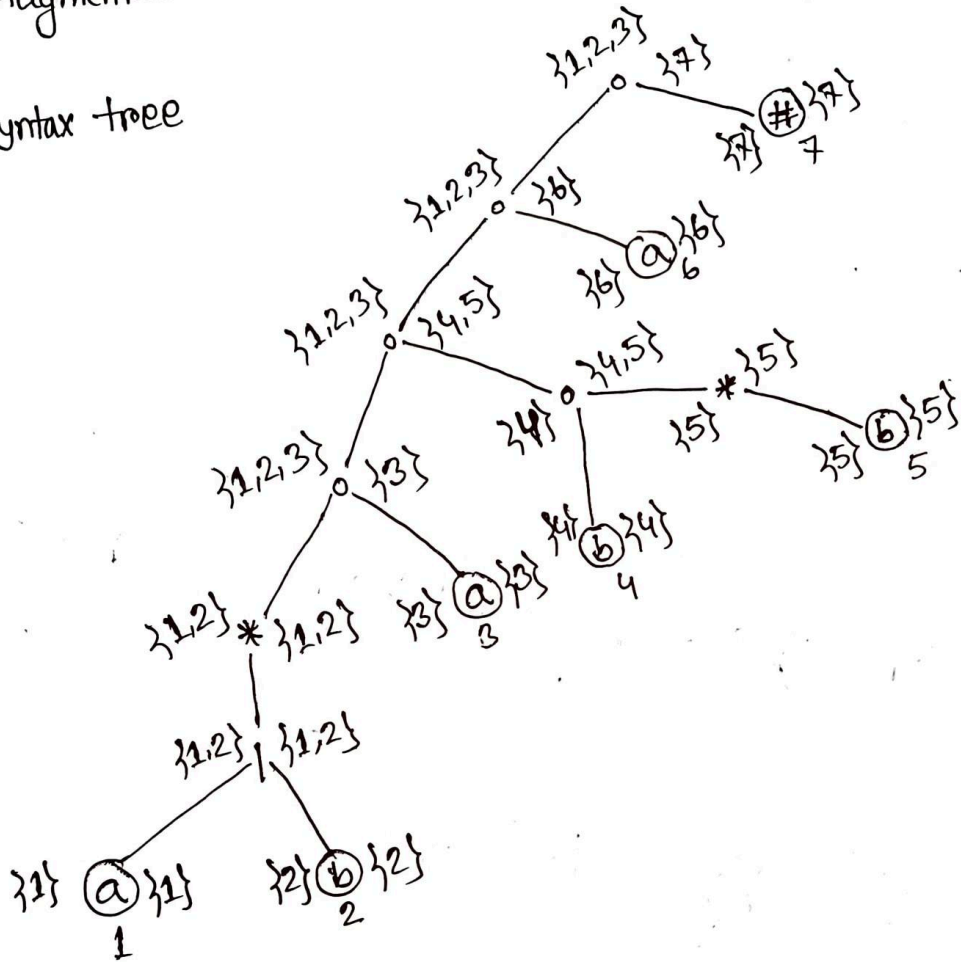
Note: You must have to show the steps.

2. Will the DFA accept any string that ends with a single **a** but no **b** before it, for example "aa"? Justify your answer briefly. [2]

① $(a|b)^* a (b|b^*) a$

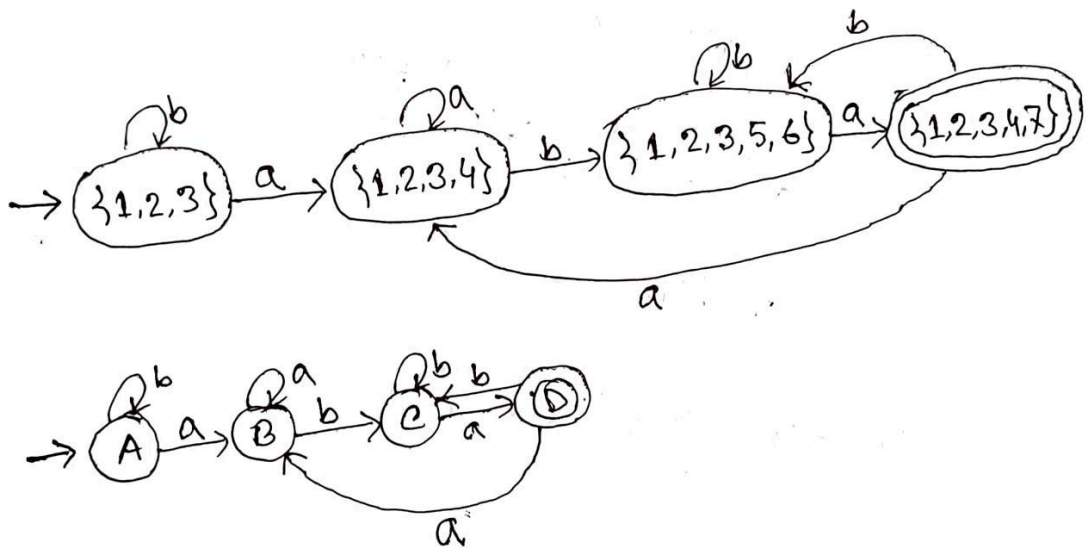
Augmented RE = $(a|b)^* \cdot a \cdot (b \cdot b^*) \cdot a$

Syntax tree



Followpos

	Node	Followpos
a	1	1,2,3
b	2	1,2,3
a	3	4
b	4	5,6
b	5	5,6
a	6	7
#	7	-



- ② No, the DFA won't accept any string that ends with a single a but no b before it. Because $(ab)^*a(bb^*)a = (ab)^*ab^+a$, any accepted string must end with one or more b 's before the final a . " aa " has no b between the two a 's, so it is rejected.